PARISHRAM 2.0

CLASS 12TH 2026

Electrostatic Potential and Capacitance

PHYSICS | LECTURE-03

BY-DEV AGNANI SIR



Topics to be covered:

1. Potential Energy in an External Field of a:
   1. Single charge
   2. System of two charges
   3. Dipole
2. Conductors and insulators } Electrostatics of Conductors.
3. Free charges and bound charges inside a conductor
4. Dielectrics and electric polarization ★★

RECAP:



1. Equipotential Surfaces

2. Relation between Electric Field and Potential

$$V = -\int \vec{E} \cdot \vec{dr}$$

3. Electrical potential energy of a system of charges



$$U = \frac{kq_1q_2}{r}$$

$$E = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$$

QUESTION

$V_r = (2x^3\,\hat{i} + 4y\hat{j} + z^2\hat{k})\ volts.$

Find the electric Field at the point (1, 1, 1).

$$E_r = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right) V/m$$

$$\Rightarrow E_r = -(6x^2\hat{i} + 4\hat{j} + 2z\hat{k})\ V/m$$

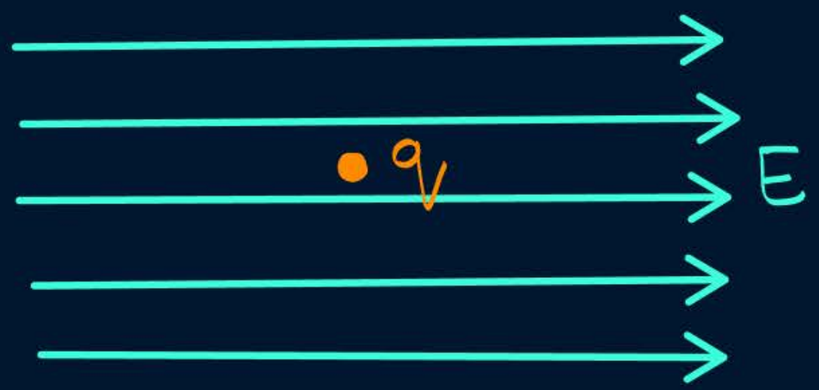
$$\Rightarrow \boxed{E_r(1,1,1) = -(6\hat{i} + 4\hat{j} + 2\hat{k})\ V/m}$$



Parishram 2.0 students when **Dev sir** scheduled a **class** on **Ganesh Chaturthi:**



Potential Energy in an External Field of a Single charge



$E \longrightarrow$ given

$V = ?$

$$V = -\int \vec{E} \cdot d\vec{r}$$

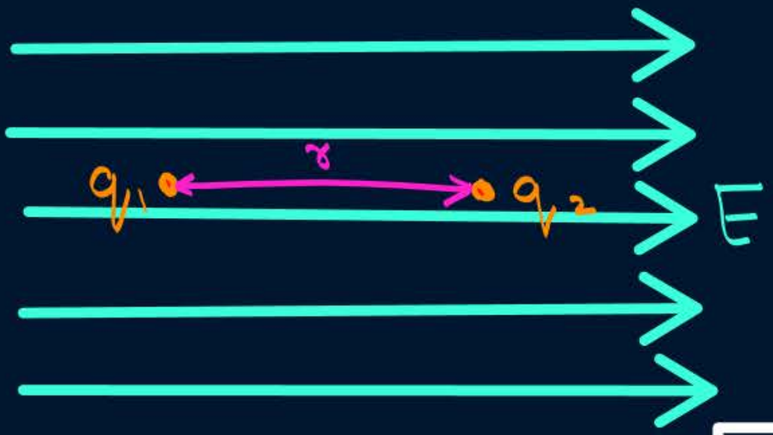
$$V = \frac{kq}{r}$$

$$U = \frac{kq_1q_2}{r}$$

$$\boxed{U = qV}$$

↳ Queen Victoria

Potential Energy in an External Field of a System of two charges



We can find V_1 and V_2 (Potentials at point 1 and point 2 respectively) by the expression:

$$V = -\int E \cdot dr$$

$$U = \underbrace{q_1V_1 + q_2V_2}_{\text{E·P·E because of External Electric Field}} + \underbrace{\frac{Kq_1q_2}{r}}_{\text{E·P·E between the two charges.}}$$

Potential Energy in an External Field of a Dipole

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$[\tau = pE\sin\theta]$$

$$W = -\int_{r_1}^{r_2} F_{ext} \cdot dr$$

Similarly

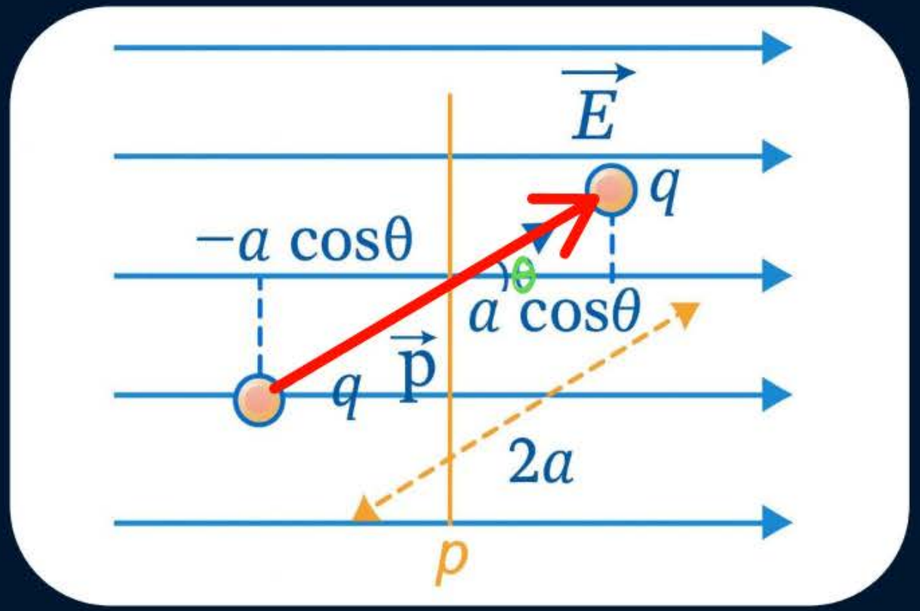
$$W = -\int_{\theta_0}^{\theta_1} \tau_{ext} \cdot d(\theta)$$

Work done to rotate a dipole from θ_0 to θ_1 in the presence of an external E.F.

$$\Rightarrow W = -\int_{\theta_0}^{\theta_1} pE\sin\theta \cdot d\theta$$

$$\Rightarrow W = -pE\left[-\cos\theta\right]_{\theta_0}^{\theta_1} = -pE\left[-\cos\theta_1 - (-\cos\theta_0)\right]$$

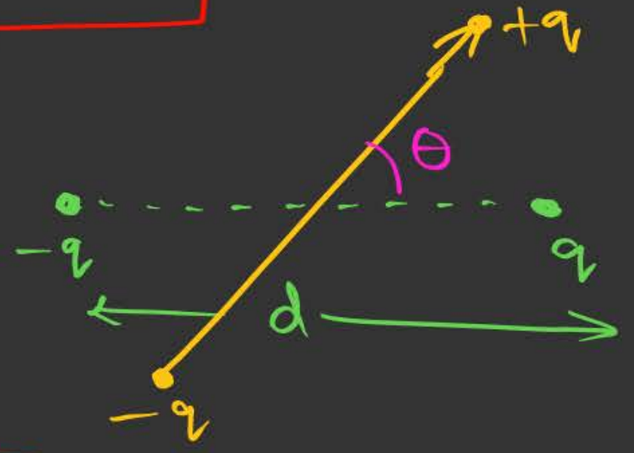
$$\Rightarrow \boxed{W = pE(\cos\theta_0 - \cos\theta_1)}$$



E.P.E. of a dipole in External E.F. ?

$$U = -pE\cos\theta$$

$$U = -\vec{P}\cdot\vec{E}$$



Special Cases

- $\theta = 180^\circ$: $U = pE$ (maximum)
- $\theta = 0^\circ$: $U = -pE$ (minimum)
- $\theta = 90^\circ$: $U = 0$

Total Electrostatic Potential Energy :-

$$U = q(V_1) - q(V_2) + \frac{Kq(-q)}{d^2} \Rightarrow U = q(V_1) - q(V_2) - \frac{Kq^2}{d^2}$$

$q(V_1) - q(V_2) = -pE\cos\theta$



QUESTION

A dipole of dipole moment 2×10^{-14} Cm is placed in a uniform electric field of 5×10^{4} N/C. Find work done in rotating the dipole from 60° to 90° ?

P

E

$W = pE(\cos\theta_0 - \cos\theta_1)$

$W = (2\times10^{-14})(5\times10^{4})\left(\frac{1}{2} - 0\right)$

$W = 2\times10^{-14}\times5\times10^{4}\times\frac{1}{2}$

$W = 5\times10^{-10}$ J

A 5×10^{-8} J

B 5×10^{-6} J

C 2.25×10^{-8} J

D 5×10^{-10} J



Example

NCERT

HW

(a) Determine the electrostatic potential energy of a system consisting of two charges 7 μC and -2 μC (and with no external field) placed at (–9 cm, 0, 0) and (9 cm, 0, 0) respectively.

(b) How much work is required to separate the two charges infinitely away from each other?

(c) Suppose that the same system of charges is now placed in an external electric field $E = A\,(1/r^2)$; $A = 9 \times 10^5\ NC^{-1}\ m^2$. What would the electrostatic energy of the configuration be?

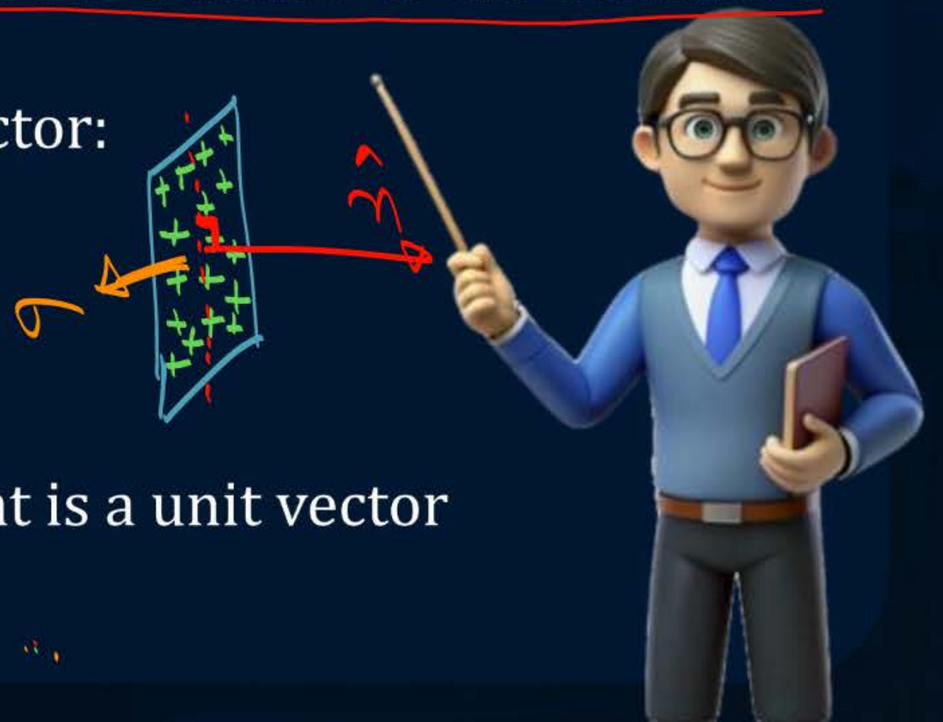


Electrostatics of Conductors

Important results regarding electrostatics of conductors:

- Inside a conductor, the electrostatic field is zero.
- At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point.
- The interior of a conductor can have no excess charge in the static situation.
- The electrostatic potential is constant throughout the volume of the conductor and has the same value (as inside) on its surface.
- The electric field at the surface of a charged conductor:

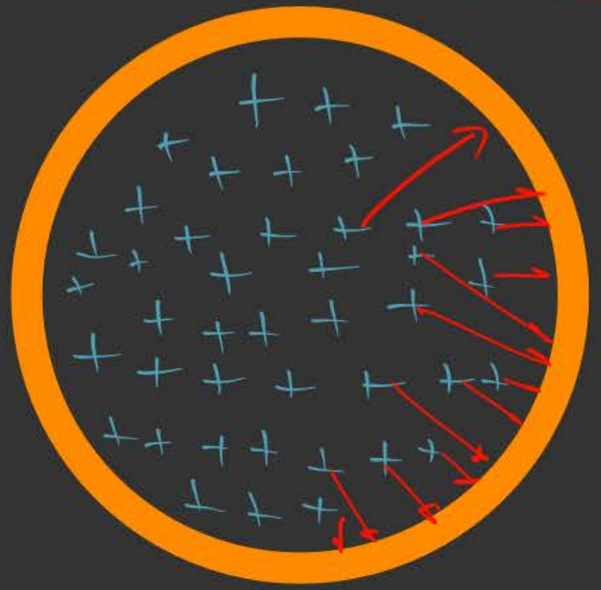
$$\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$$

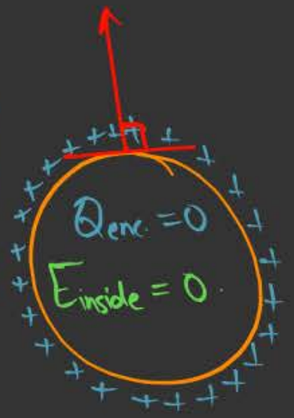


where sigma is the surface charge density and n-hat is a unit vector normal to the surface in the outward direction.

CONDUCTOR

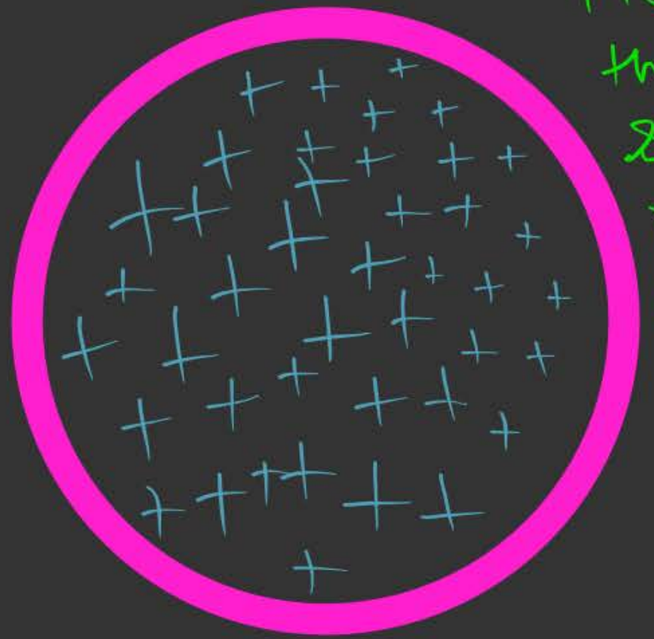
$V_{inside} = V_{surface}$





All the charges placed inside the conducting sphere will move to its surface in order to gain stability.
Because these are free charges.

Non-Conductor
(insulator).



Here, charge inside the non-conducting sphere will not move, because these are not free charges.

Leakage of Charge

Egg? → Conductor → L
↳ Non-Conductor → DL

Leakage of charge from sharp points of a conductor.

Here $V_A = V_B$ but $\sigma_B > \sigma_A$ and here $E_B > E_A$

$$\left\{\therefore E = \frac{\sigma}{\varepsilon_0}\right\}$$

$Q_B > Q_A$



Read.

Thus charge density and here electric field is larger at sharp corners as compared to flat surfaces due to their less radius of curvature.

$$\sigma_1 = \frac{Q_1}{4\pi R_1^2} \text{ and } \sigma_2 = \frac{Q_2}{4\pi R_2^2}$$

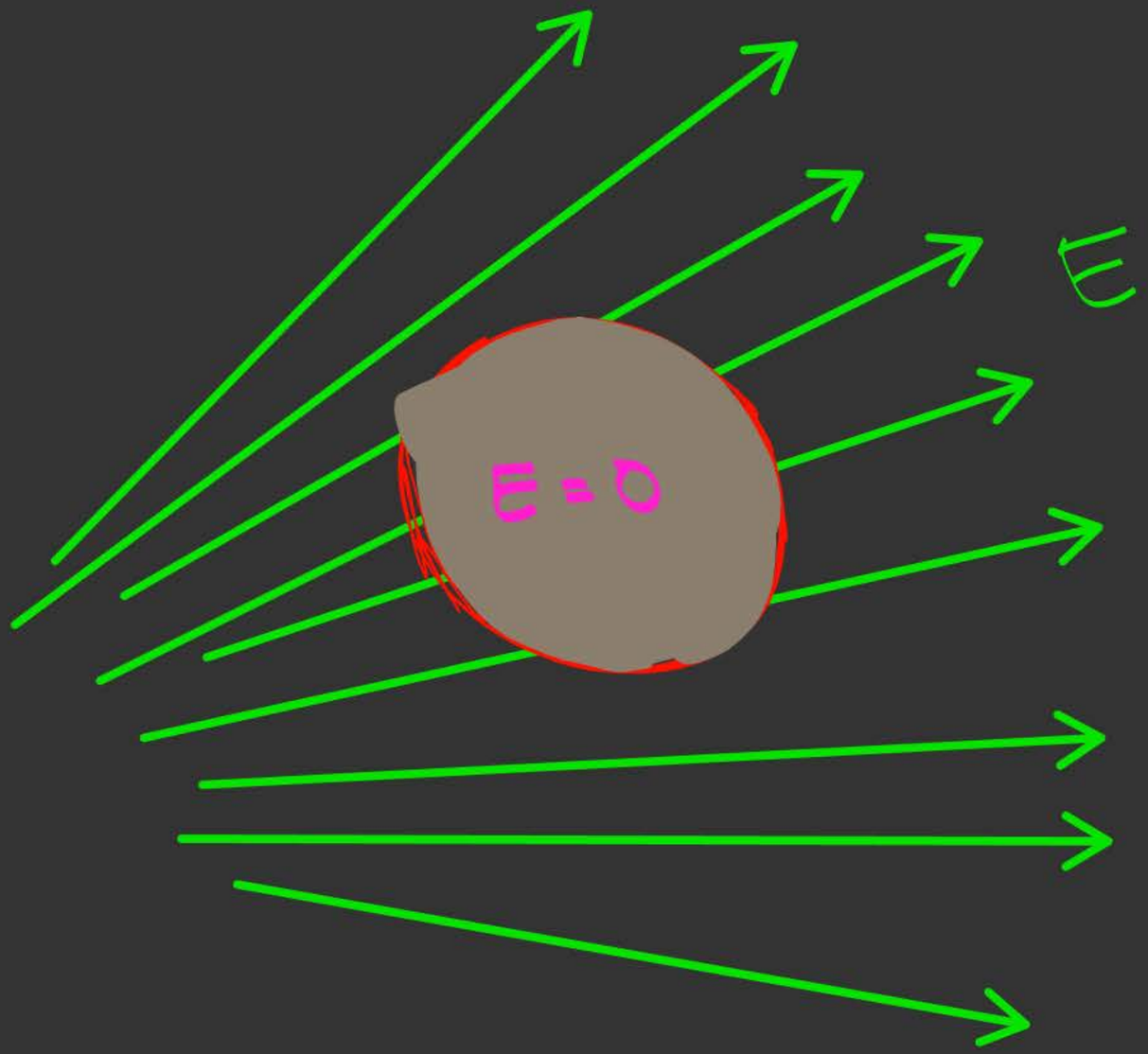
$$V = \frac{KQ_1}{R_1} = \frac{KQ_2}{R_2} \Rightarrow \frac{Q_1}{Q_2} = \frac{R_1}{R_2}$$



Electrostatic Shielding

- Protecting a specific area from the effects of an electric field can be done by electrostatic shielding.
- When conducting walls enclose a cavity, the cavity is considered field free as long as no charges are present within.
- Whatever be the charge and field configuration, the field inside the cavity is always zero (Provided no charge is present inside the cavity). This is known as **electrostatic shielding.**
- One way to accomplish electrostatic shielding is to place sensitive instruments within a hollow conductor.



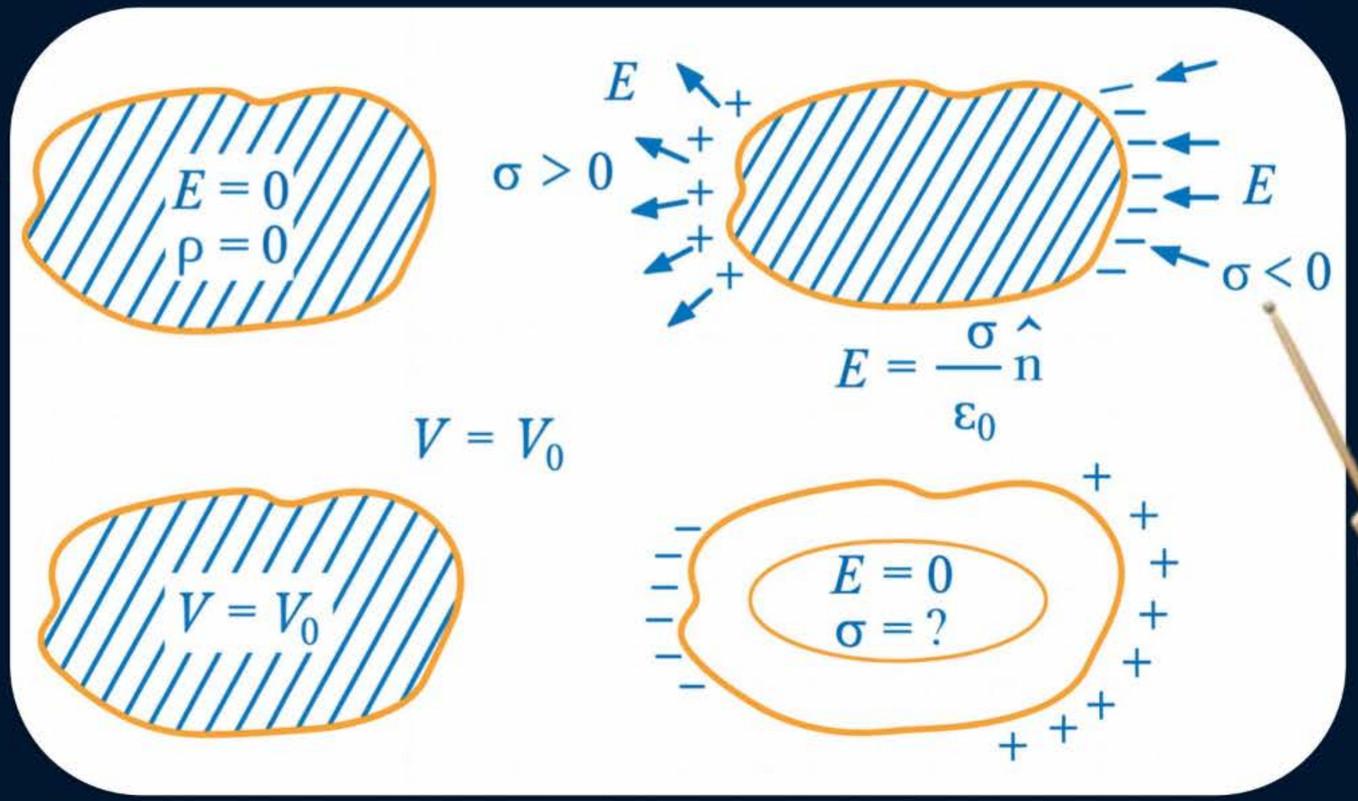


To protect this charge q from the influence of the $\vec{E}$, we will do the Electrostatic shielding of this charge.

Electrostatic Shielding

❖ Figure gives a summary of the important electrostatic properties of a conductor.)

Potential will exist inside the cavity even after the Electrostatic shielding is done.



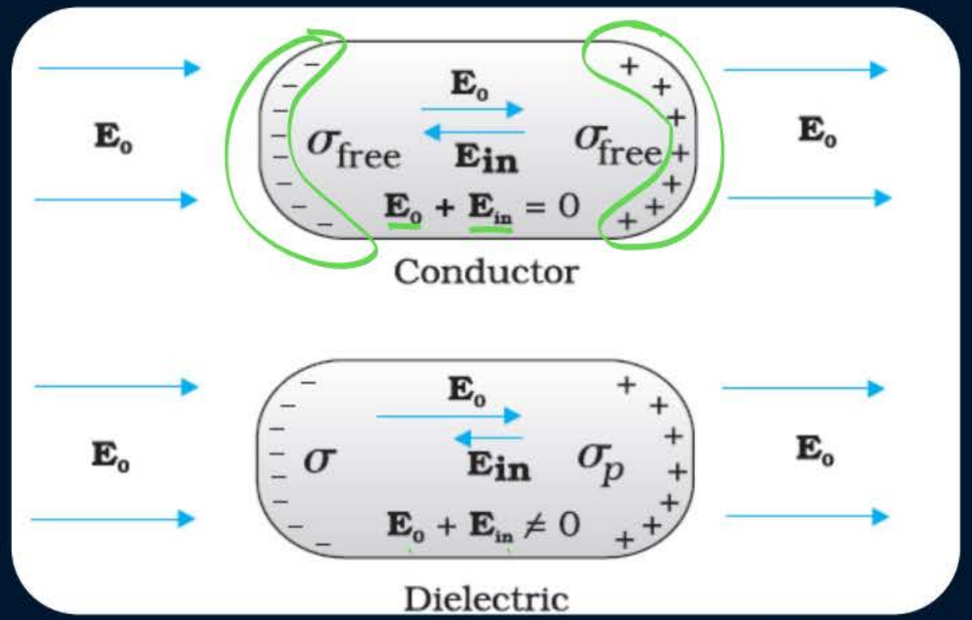


Dielectric

$F = \frac{kq_1q_2}{r^2}$ → k → Electrostatic Constant or Coulomb's Constant.

Notes

- A **dielectric** is a non-conducting substance (insulator) which can transmit electric effects without conducting current.
- In other words, it does not **allow free flow of electric charge,** but it gets **polarized** when placed in an electric field.
- **Example:** glass, mica, plastic, porcelain, distilled water, wood.



$K = \frac{1}{4\pi\epsilon_0}$ → air.

medium → m.

Dielectric Constant = $K = \epsilon_r = \frac{\epsilon_m}{\epsilon_0}$

ϵ_m → Permittivity of the medium

ϵ_0 → Permittivity of the air/vacuum.

OR

Relative Permittivity.

∴

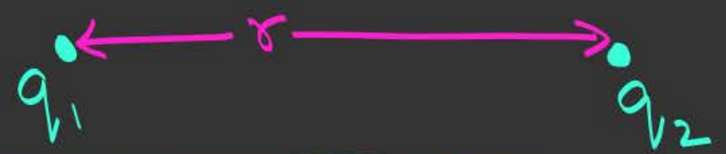
Air



$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}$$

Medium → K

Dielectric constant of the medium.



$$F = \frac{1}{4\pi K\epsilon_0}\frac{q_1 q_2}{r^2}$$

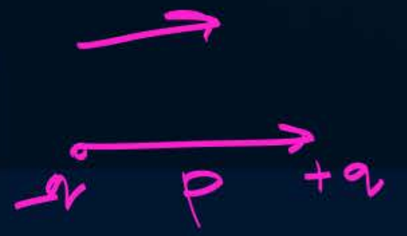
$$K = \epsilon_r = \frac{\epsilon_m}{\epsilon_0}$$

$$\Rightarrow \epsilon_m = K\epsilon_0$$

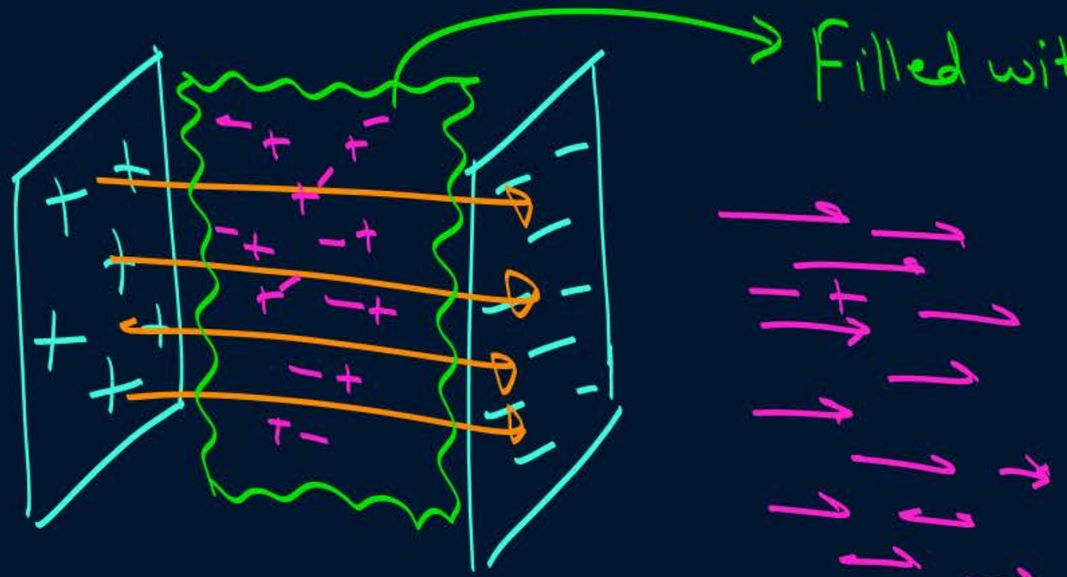
or

$$\epsilon_m = \epsilon_r\epsilon_0$$

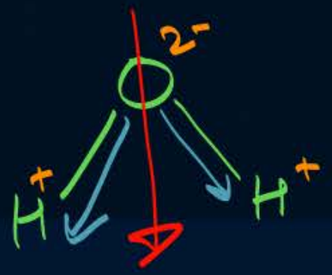
Behavior of Dielectric in Electric Field



- In the absence of an electric field → molecules of dielectric have **random orientation** of charges.
- When placed in an external electric field → charges **slightly shift** from their mean positions, and molecules align → the dielectric becomes **polarized**.
- This polarization reduces the **effective electric field** inside the dielectric.



Types of Dielectrics





Notes

Polar Dielectric

- Molecules have **a permanent dipole** moment due to unequal charge distribution.
- In absence of electric field → dipoles are randomly oriented
  (Net Effect = 0).
- In presence of electric field → dipoles align partially along the field.
- **Examples:**
  Water (H_2O), HCl, NH_3.

Non-Polar Dielectric

- Molecules have **no permanent dipole** moment.
- In absence of electric field → charge distribution is symmetric.
- In presence of electric field → temporary dipole is induced (Induced Polarization).
- **Examples:**
  Oxygen (O_2), Nitrogen (N_2), CO_2, H_2, benzene.

Dielectric Constant (Relative Permittivity, K)

Notes

- Define as:

$$K = \frac{\text{Permittivity of Dielectric } (\varepsilon)}{\text{Permittivity of Free Space } (\varepsilon_0)}$$

- $K > 1$ | $K = 1$ for air/vacuum/free space.
- Higher K → better dielectric property → Increases capacitance of a capacitor.

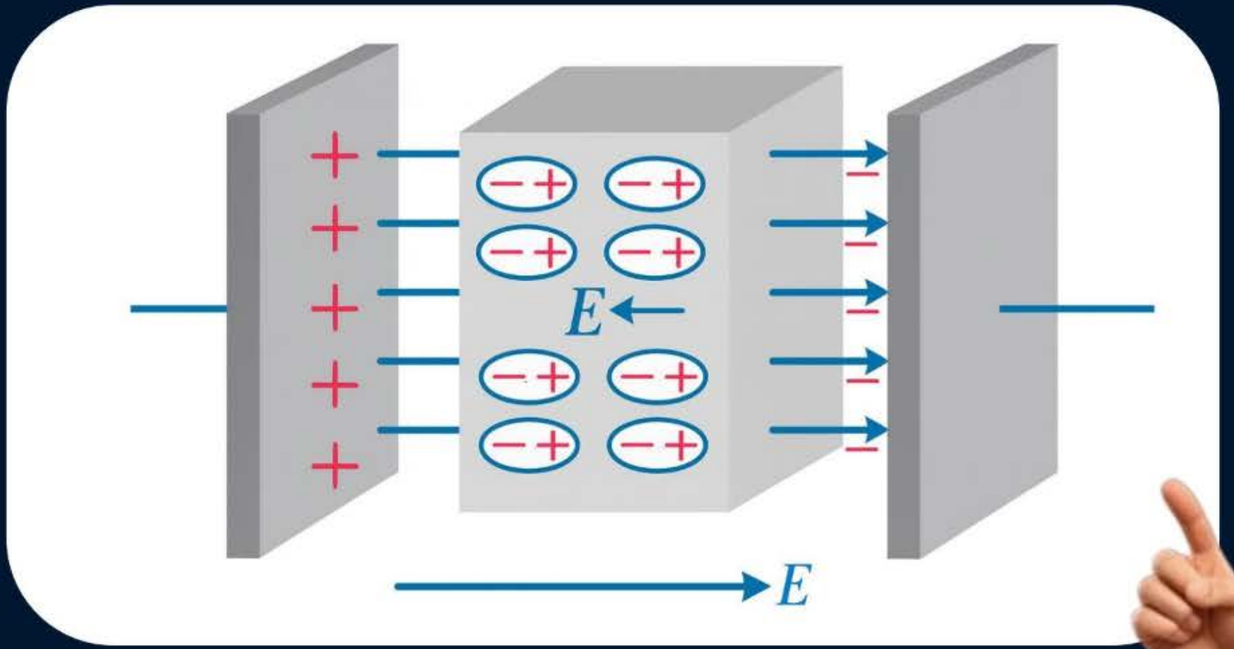
- Dielectric constant of a dielectric medium is defined as:

$$\frac{E}{E'} = \frac{\text{Electric Field between the Plates in Vaccum/Air}}{\text{Electric Field between the Plates with Medium}} = K$$

Polarization

Revision Notes

Polarization of a Dielectric Slab: It is the process of inducing equal and opposite charges on the two faces of the dielectric on the application of **electric field**.



Electric field between the plates in the presence of dielectric medium is **$E' = E - E_i$**, where **E = external electric field, E' = Induced field.**



Susceptibility

The dipole moment per unit volume is called **polarisation** and is denoted by **P**. For linear isotropic dielectrics,

$$\vec{P} = \varepsilon_0 \chi_e \vec{E}$$

Polarisation — Permittivity — Susceptibility — Electric field.

Where χ_e is a constant characteristic of the **dielectric** and known as the **electric susceptibility** of the dielectric medium.



QUESTION

A dielectric material is always:

non-conductor

A. A conductor

B. An insulator

C. A semiconductor

D. A superconductor



QUESTION

When a dielectric is placed between the plates of a charged capacitor, the potential difference between plates:

A Increases

B Decreases

C Remains same

D Becomes zero



QUESTION

Dielectrics

Which of the following is a **polar dielectric**?

A CO_2 O=C=O

B H_2 H—H

C HCl $H^+ \rightarrow Cl^-$

D N_2 N—N



QUESTION

The dielectric constant of vacuum is:

$K = \epsilon_r = \frac{\epsilon_m}{\epsilon_0}$

$K = \frac{\epsilon_0}{\epsilon_0} \Rightarrow \boxed{K = 1}$

A Zero

B One

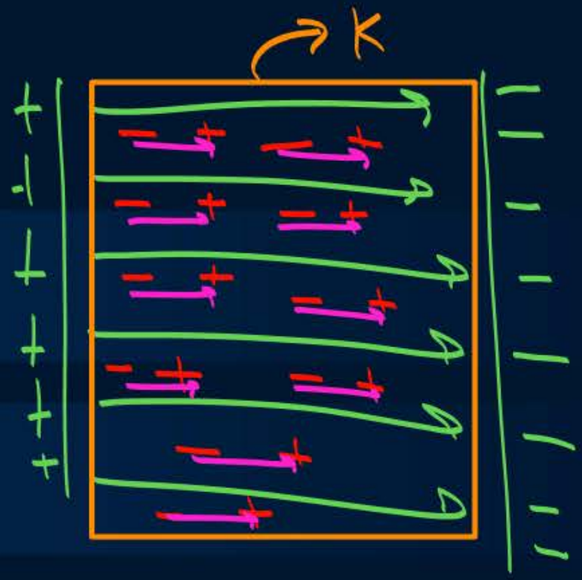
C Infinite

D Depends on conditions



QUESTION

In a non-polar dielectric, polarization is due to:



A Permanent dipole moment

B Thermal vibration

C Induced dipole moment

D Free electrons



AI Board Booster: **Student se Super Student**

Ek aisa course jo **AI ke use** se aapki Board prep level-up kare

Outcomes- Aapko kya milega? (padhai mein badlao)

- **Faster revision:** Turn chapters into crisp **summaries, flashcards, and quizzes** in minutes.
- **Target weak areas:** AI **flags your common mistakes** and suggests exact topics to revisit.
- **Exam-style practice:** Convert **past papers into timed mocks** with stepwise solution.
- **Showcase work:** Build **8–10 mini projects** (notes generator, quiz bot, image-to-notes, etc.).
- **Repeatable study systems:** Unit tests & Boards ke liye **plug-and-play** checklists/templates.
- **Responsible AI use:** Kya **allowed** hai, kaise cite karna hai, and **over-reliance** avoid karna
- **Real-world edge: Prompting & automation** basics jo college/internships mein use karoge.
- **Recognized proof: PW Skills certificate** *(signed by Founder Alakh Pandey).*

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For classes 9–12 (all boards)



Course at a glance

- **Live Online | Hinglish | Mentor:** Vishal Tiwari Sir
- **18 sessions × 3 hrs (Saturdays)**
- **Weekly Doubt Room:** Mon, 45–60 min
- **Start:** 13 Sept 2025
- **Fee:** ₹2500 → ₹799 for Uday & Parishram paid users *(coupon in tray)*

Limited seats - Offer ends soon

DPP के Ques. को अपने Notes में as Subjective Questions लिखना चालू करदो।

THANK YOU

